UNIVERSITAT
ROVIRA i VIRGILI

Artificial Intelligence Seminar

Design and implementation of a system for the evaluation of the effectiveness of a recommender system

---

Altés Grifoll, Biel

Universitat Rovira i Virgili

Contents

1 Introduction

Barcelona is one of the clearest examples of what happens when tourism grows faster than a city can handle. The city received about 14.5 million visitors in 2024, and a small group of famous places receives most of that pressure. The Sagrada Família alone reported 4,707,367 visitors in 2023, and Park Güell closed the same year with around 4.4 million [1], [2]. Tourist accommodation is also shared in a very unequal way. Two central districts, Ciutat Vella and the Eixample, together have more than half of the city's hotel and guesthouse beds, while outer districts such as Nou Barris and Sant Andreu have less than half a percent each [3]. The result is the well-known problem of overtourism, crowded streets around a few famous places, higher costs for residents, and whole districts that tourists almost never visit.

A recommender system can make this problem worse, or it can help to solve it. A system that always suggests the most popular places sends even more people to the same few spots. The seminar discussed a different idea, a multi-criteria recommender that also thinks about sustainability. This kind of system would still respect what the visitor wants, but it would also consider social, cultural, economic and environmental factors, so that tourists are spread more evenly and less famous places also get some attention. The points-of-interest seminar [4] explained this in a practical way, and connected the idea to methods such as sending tourists to quieter attractions and following how they move through the city.

This work does not build that recommender. Instead, it asks a question that is easy to state but surprisingly hard to answer. If such a system already existed, how would we know that it really improves the management of tourists, instead of just assuming that it does? A recommender can look good in theory and still fail to change behaviour when real people react to crowds, distances and budgets. To study this, we built an agent-based simulator. The simulator creates a large population of tourists, gives each one a profile, lets several recommenders suggest places, and then simulates which places the tourists really visit. From those visits we measure the crowding at each point of interest, how tourists are spread across neighbourhoods, and a set of quality indicators for the recommenders. The main comparison is between a sustainability-aware recommender and three simpler baselines.

So the contribution of this report is an evaluation method, not a new recommender. We describe a dataset of 77 real points of interest in Barcelona that anyone can reproduce, four recommendation strategies, an agent-based model written with the Mesa framework, and a set of metrics based on the seminar material. We then report what the simulation shows. In short, the sustainability-aware recommender does reduce overcrowding and does spread visitors more evenly across districts, but the price is a clear drop in how well it matches each visitor's stated interests. The rest of the report explains how we reached this conclusion and how much we can trust it.

2 Background

2.1 Recommender systems and their evaluation

A recommender system filters a large set of items and proposes the ones that seem most relevant to a given user. The introductory seminar [5] separated accuracy from what we often call beyond-accuracy properties. Accuracy metrics such as precision, recall and normalised discounted cumulative gain ask whether the system found the items the user would have chosen. Beyond-accuracy metrics look at the things that are still important after we cover these obvious matches. Diversity measures how different the recommended items are from one another, novelty rewards a system that is ready to suggest less popular items, and coverage measures how much of the full catalogue is recommended at all. These properties are important here because a sustainability goal is mostly a beyond-accuracy goal, spreading tourists out is the opposite of always recommending the same famous places.

2.2 Tourism recommenders and overtourism

Tourism is a difficult case for recommenders because the items are real places with limited capacity, opening hours and a fixed location. The points-of-interest seminar explained the sustainability problem in a practical way [4]. It described three things that a tourism recommender can do, send visitors to less crowded attractions, follow how people move so that the city can react, and control which attractions are shown to which visitors. All three ideas assume that the recommendations really change where people go, and this is exactly what a simulation can test.

2.3 Multi-criteria decision aiding

The sustainability-aware recommender is based on multi-criteria decision aiding. The multi-criteria seminar [6] described a recommendation as a ranking problem over several criteria that we cannot compare directly, for example a visitor's interest, the crowding of a place, and its environmental quality. The seminar presented the ELECTRE family of outranking methods, which compare options in pairs using concordance and discordance, together with a simpler weighted average. The related sustainable-tourism material organised the criteria into four groups that cover environmental, cultural, social and economic topics. We follow this structure when we build the criteria for our recommender, and we use ELECTRE-III as the ranking method because this was the method presented in that seminar.

2.4 Context, fairness and trust

Three more seminars influenced the way we evaluate the system. The context-aware seminar [7] described how factors such as the group, the budget and the available time change what makes a good recommendation, and this is the reason for the detailed tourist profiles we create. The trustworthy systems seminar [8] listed properties such as transparency, explainability and fairness. It made a point that is very important for our design, offline evaluation misses the feedback loop that appears when real recommendations change behaviour, and this behaviour then changes the future data. An agent-based simulation is one way to bring that loop back. We make the fairness property practical by treating it as a question of exposure, where some items are shown much more often than others, and we measure it with a Gini coefficient of how often each place is recommended. Finally, the knowledge-graph seminar [9] argued that recommendations should be based on clear reasoning over a knowledge graph, and paid attention to the popularity and diversity of the explanations a system gives. We return to this idea later, when we read our own recommendations as per-criterion explanations.

3 Data

3.1 Sources and integrity

We built a dataset of 77 points of interest in Barcelona. Every field with data comes from a source that we can cite, and any field without a reliable source was left empty instead of guessed. Names, coordinates and heritage status come from Wikidata [10]. The popularity of each place is based on its Wikipedia page views over the twelve months from June 2025 to May 2026. For accessibility and price tags we used OpenStreetMap [11]. District and neighbourhood labels follow the official city maps, we assign each place to the area that contains its coordinates (a point-in-polygon method). The district areas come from the 2024 statistical yearbook [12], and the visitor numbers for a few major attractions come from each site's own reports and from city press releases [1], [2], [13].

3.2 Schema and derived criteria

Each point of interest has a category from a fixed list, its coordinates, its district and neighbourhood, a popularity value, an entrance fee, a green-space flag, an accessibility label and a heritage-protection level. From these basic fields we compute the criteria that the recommenders use. Popularity is the page-view total rescaled to the range from zero to one. The environmental score is simply the green-space flag. Accessibility turns the OpenStreetMap labels into a number, and cultural value turns the Catalan heritage levels (national, local, or none) into its own score.

Two fields need a clear explanation, because the data made us take some decisions. First, no point of interest publishes an official daily visitor limit, so the capacity field is empty for all of them. For this reason we use the same fixed capacity for every place, and we keep it on purpose independent of popularity, so that crowding comes from demand and not from the rule itself. Second, there is no official number of tourists per square kilometre for each district. We replace it with a value that we can really find, the district's share of the city's tourist beds divided by its area [3], [12]. This puts Ciutat Vella above the Eixample, which matches the known pattern of pressure in the city. From this we define a low-pressure score, higher for calmer districts, and a local-economy score based on the distance of each place from the central tourist area around Plaça de Catalunya. Both choices are simplifications, and we discuss them again in the limitations.

4 System design

4.1 Tourist profiles

A profile describes a single tourist. It has a weight for each category of interest, a budget for entrance fees, a transport mode (walking or public transport), a tolerance for walking, a crowd aversion, a sustainability sensitivity, a preference for outdoor places, information about whether the group travels with children, the group size, the planned number of visits, and the district where the trip starts. The population generator creates these values with a fixed seed, so we can reuse the same population with every recommender. Most tourists care about one to three categories more than the others, which creates the kind of varied demand that a real city has.

4.2 The four recommenders

All four recommenders use the same interface, each one receives a tourist profile and the set of places, and returns a ranked list of place identifiers.

The popularity recommender ranks places by their page-view popularity and ignores the profile completely. It is the strategy that a simple system would use, and it is the one that most probably sends everyone to the same famous places.

The interests recommender ranks places by how well their category matches the profile, after it removes the places whose fee is higher than the budget. It is a strong personalised baseline that should get good accuracy.

The sustainability-aware recommender is the system that we test. It scores each place on six criteria, the match with the visitor's interests, the low-pressure score of its district, its environmental score, its accessibility, its cultural value, and its local-economy score. The weight between preference and sustainability depends on the visitor. If s is the sustainability sensitivity of the profile, the preference criterion receives weight $1 - 0.6\,s$, and the five sustainability criteria share the rest of the weight equally, so each one receives $0.6\,s/5$. A visitor with no sustainability sensitivity is ranked almost only on interest, while a very sensitive visitor still keeps an important part of the weight on preference. We then rank the places with ELECTRE-III, using the implementation in the pyDecision library [14], after a fast weighted-average pre-selection that keeps the method quick on the full catalogue. We keep a weighted-average ranking as a check, which also matches the simpler method shown in the multi-criteria seminar [6].

The random recommender returns the places in a random order. It works as a lower bound that helps us understand every other number in the comparison.

4.3 The agent-based model

The simulation is built on the Mesa framework [15], and each tourist is modelled as a single agent. At the start of the trip, the agent asks the active recommender for its ranked list one time. This is like a real visitor who receives a set of suggestions and then visits them during the day. At each step, the agent looks at the next few suggestions that it has not visited and can afford, and chooses one of them. This choice is not completely greedy. The agent balances the rank of the place, its current crowding, and the travel cost from its current location. Crowding has more weight when the agent has a higher crowd aversion, and travel has more weight when the agent has a lower walking tolerance or uses a different transport mode. The agent then visits the chosen place, and this updates its budget, its location, and the crowding counters of the place and its district.

The city model keeps the shared state of the simulation, with a current count of visits at every place and in every district, and a record of each place's capacity. When a place goes over its capacity, it becomes less attractive, so the crowd-averse tourists who come later usually go to other places. This is the feedback loop that the trustworthy systems seminar said is missing from static evaluation [8]. We run every strategy on the same population with the same seed, so the only thing that changes between runs is the recommender.

4.4 Interactive demo

The repository also includes an interactive dashboard built with Streamlit, which turns the results of the simulation into something that a reader can explore directly. The dashboard draws the points of interest on a map of Barcelona, where the size and colour of each marker grow with how crowded that place becomes under the chosen recommender. When you change the recommender, the map updates, so the difference between the popularity strategy and the sustainability-aware one is easy to see. The same view shows the metric comparisons from this report as charts, together with the division of visits across the ten districts. The dashboard runs on your computer with the command `streamlit run demo/streamlit_app.py`, and you can publish it as a web application in the same way as any other Streamlit project. Figures 1 and 2 show the dashboard during use.

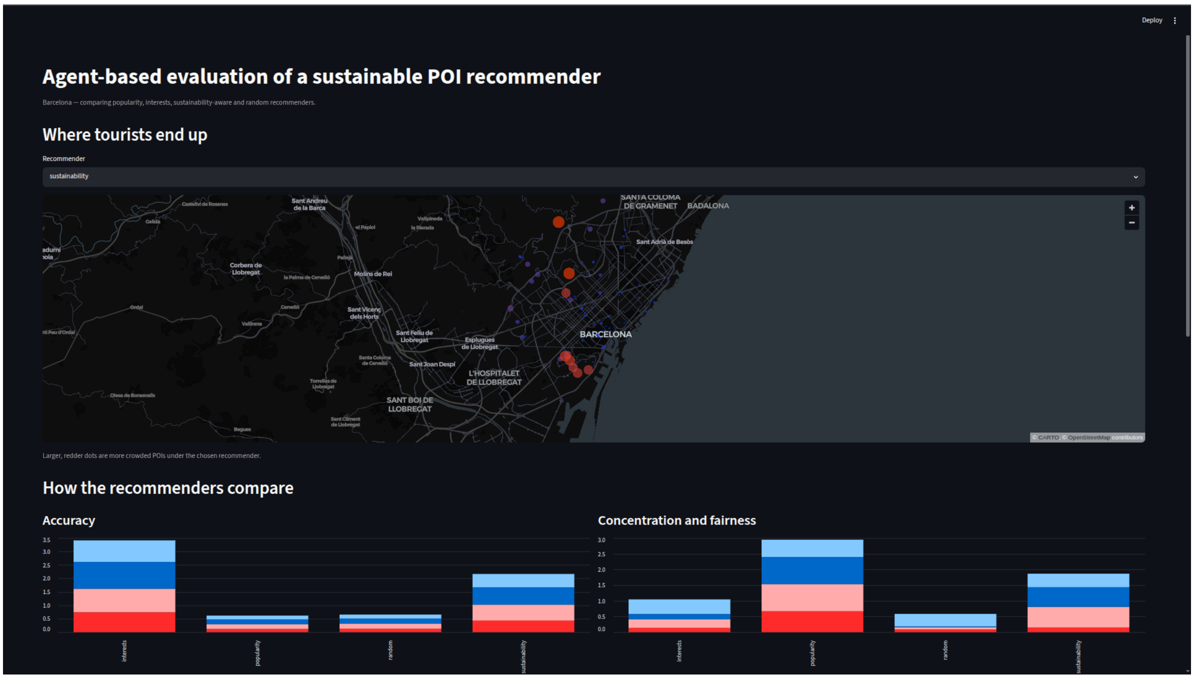

Figure 1: The Streamlit dashboard, with the crowding map of Barcelona for the selected recommender and the first metric comparisons below it.

strategy	precision	recall	f1	ndcg	diversity	novelty	coverage	personalisation	poi_visit_gini	district_visit_gini	exposure_gini	overcrowded_pois	top_popular_visit_share	visited_low_pressure	visited_environmental	visited_accessibility	visited_cultural	visited_local_economy	low_pressure_district_share
popularity	0.178	0.117	0.141	0.181	0.535	4.581	0.13	0	0.862	0.552	0.87	9	0.667	0.535	0.081	0.798	0.815	0.275	0.235
interests	0.878	0.738	0.802	0.998	0.333	9.052	0.961	0.861	0.277	0.472	0.162	0	0.135	0.517	0.119	0.689	0.583	0.308	0.317
sustainability	0.591	0.431	0.499	0.645	0.526	8.62	0.974	0.665	0.658	0.432	0.626	5	0.151	0.86	0.254	0.754	0.728	0.473	0.641
random	0.181	0.131	0.152	0.191	0.606	9.563	1	0.872	0.07	0.404	0.02	0	0.088	0.558	0.136	0.661	0.561	0.334	0.336

Figure 2: The lower part of the dashboard, with the diversity, coverage and sustainability charts, the division of visits across the ten districts, and the full metrics table.

5 Experimental setup

The simulation runs with a population of 5,000 tourists, and each tourist plans between three and eight visits, so a single run produces about twenty-seven thousand visits over the 77 places. We run the four recommenders on this single population. For each one, we record the full recommendation list of every tourist, the places that each tourist actually visited, and the final crowding state of the city.

The metrics fall into four groups, and we describe each group one by one. The first group, accuracy, uses precision, recall, the F1 score and normalised discounted cumulative gain at a cut-off of ten. A place is relevant to a tourist when it belongs to a category that the tourist clearly likes, this follows the category-based relevance used in the points-of-interest seminar [4]. The beyond-accuracy group uses intra-list diversity, novelty and catalogue coverage. Intra-list diversity is the average difference between pairs of recommended places, and it combines a category difference and a geographic distance. Novelty is the average of $-\log_2 p_i$ over the recommended places, where p_i is the normalised popularity of place i, so places that few people visit get a higher score. Coverage is the part of the catalogue that appears in any recommendation.

The third group measures fairness and concentration. Here we use the Gini coefficient of the visit counts across places, the Gini of visits across districts, and the Gini of how often each place is recommended, which sees fairness as a question of exposure. We also count how many places go over their capacity, and we report the share of all visits that go to the most popular tenth of the catalogue. The fourth group measures the sustainability of the visits that really happened. For each sustainability criterion, we compute its average over the visited places, weighted by the number of visits, and we report the share of visits that reach low-pressure districts.

6 Results

Table 1 shows the accuracy and beyond-accuracy metrics, and Table 2 shows the fairness, concentration and sustainability metrics. All the numbers come from the same 5,000-tourist run.

Table 1: Accuracy and beyond-accuracy metrics by recommender.

Metric	Popularity	Interests	Sustainability	Random
Precision@10	0.178	**0.878**	0.591	0.181
Recall@10	0.117	**0.738**	0.431	0.131
F1@10	0.141	**0.802**	0.499	0.152
NDCG@10	0.181	**0.998**	0.645	0.191
Diversity (ILD)	0.535	0.333	0.526	**0.606**
Novelty	4.58	9.05	8.62	**9.56**
Coverage	0.130	0.961	0.974	**1.000**
Personalisation	0.000	0.861	0.665	**0.872**

Table 2: Fairness, concentration and sustainability results by recommender. Lower is better for the Gini rows, the overcrowding count and the top-popular share. Higher is better for the visited-criteria rows.

Metric	Popularity	Interests	Sustainability	Random
POI visit Gini	0.862	0.277	0.658	**0.070**
District visit Gini	0.552	0.472	0.432	**0.404**
Exposure Gini	0.870	0.162	0.626	**0.020**
Overcrowded POIs	9	**0**	5	**0**
Top-popular visit share	0.667	0.135	0.151	**0.088**
Visited low-pressure	0.535	0.517	**0.860**	0.558
Visited environmental	0.081	0.119	**0.254**	0.136
Visited accessibility	0.798	0.689	0.754	0.661
Visited cultural	0.815	0.583	0.728	0.561
Visited local-economy	0.275	0.308	**0.473**	0.334
Low-pressure district share	0.235	0.317	**0.641**	0.336

6.1 Accuracy

The accuracy results are the least surprising part of the study, and Figure 3 shows them. The interests recommender is the best on every accuracy metric, with a precision at ten of 0.88 and an almost perfect ranking quality of 1.00. This is normal, because that recommender optimises directly for category match. The sustainability-aware recommender is in the middle, with a precision of 0.59 and a ranking quality of 0.65. The popularity and random strategies are much lower, both near 0.18, because matching the average famous place to the interests of one specific visitor is mostly luck. The message here is quite clear, adding sustainability criteria does reduce accuracy, but the sustainability-aware recommender still keeps a clear and useful link to what each visitor wants, and it stays well above the popularity baseline that many real systems use.

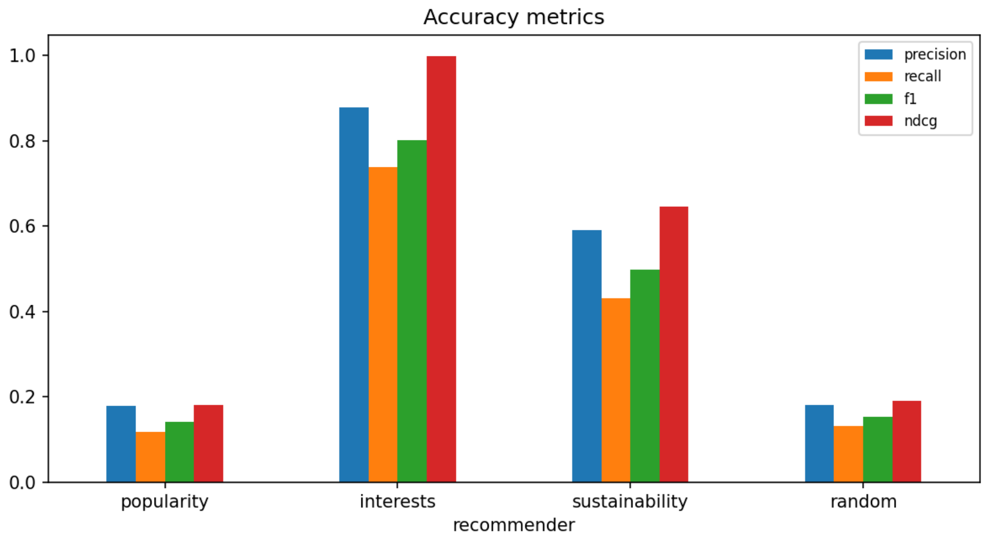

Figure 3: Accuracy metrics by recommender. The interests recommender is first, the sustainability-aware recommender is second, and popularity and random are last.

6.2 Concentration and overcrowding

Concentration is where the sustainability idea starts to show its value, as Figure 4 shows. With the popularity recommender, two thirds of all visits go to the most popular tenth of the places, the visit Gini reaches 0.86, and nine places go over their capacity. The sustainability-aware recommender changes this situation, the top-popular share falls from 0.67 to 0.15, and the number of overcrowded places drops from nine to five. The interests recommender also reduces concentration, simply because varied interests spread people across many places. The important point is that the popularity strategy, which is the easy default option, is clearly the worst for crowding, and the sustainability-aware strategy is the one designed to solve this problem.

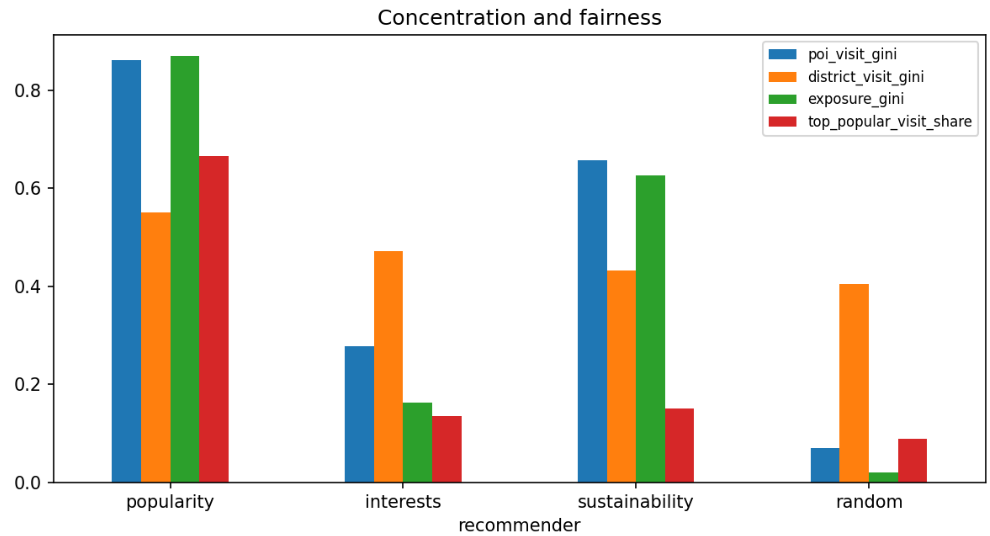

Figure 4: Concentration and fairness by recommender. Popularity concentrates both visits and exposure, the sustainability-aware recommender reduces both, but not as much as the interests baseline.

There is a real conflict in these numbers, and we should explain it. The sustainability-aware recommender has a higher place-level visit Gini (0.66) than the interests recommender (0.28). This looks contradictory at first, but it is not, when we read it carefully. The recommender sends visitors on purpose to a small chosen group of sustainable places, so visits do build up on those particular places, even though the spread across the whole city still improves. The district results, which we look at next, confirm this wider geographic spread.

6.3 Distribution across the city

The clearest sustainability benefit is about location, and it appears in both the district metrics and the maps. The sustainability-aware recommender sends 64 percent of visits to low-pressure districts, compared with only 23 percent for the popularity recommender and 32 percent for the interests recommender. Its district visit Gini, 0.43, is lower than both popularity (0.55) and interests (0.47). Figure 5 shows how each strategy divides visits across the ten districts, and Figure 6 shows the same effect on the map. With popularity, a few central places become very busy while the rest of the city stays empty. With the sustainability-aware recommender, the large markers move outward, toward Montjuïc, the upper districts and the outer areas. This is the behaviour that the city would want from a system designed to reduce pressure in the centre.

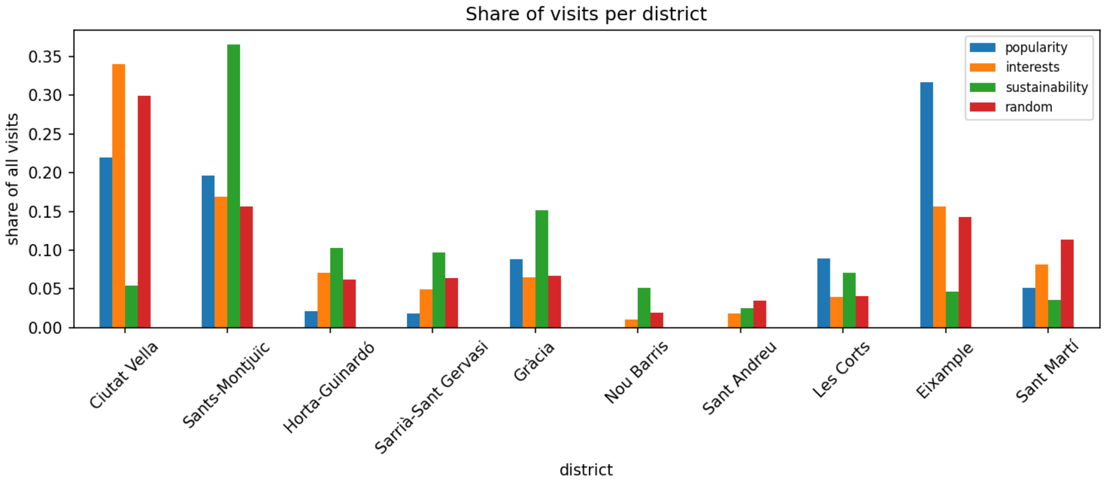

Figure 5: Share of visits per district. The popularity recommender concentrates visits in a few central districts, while the sustainability-aware recommender sends a larger share to calmer districts.

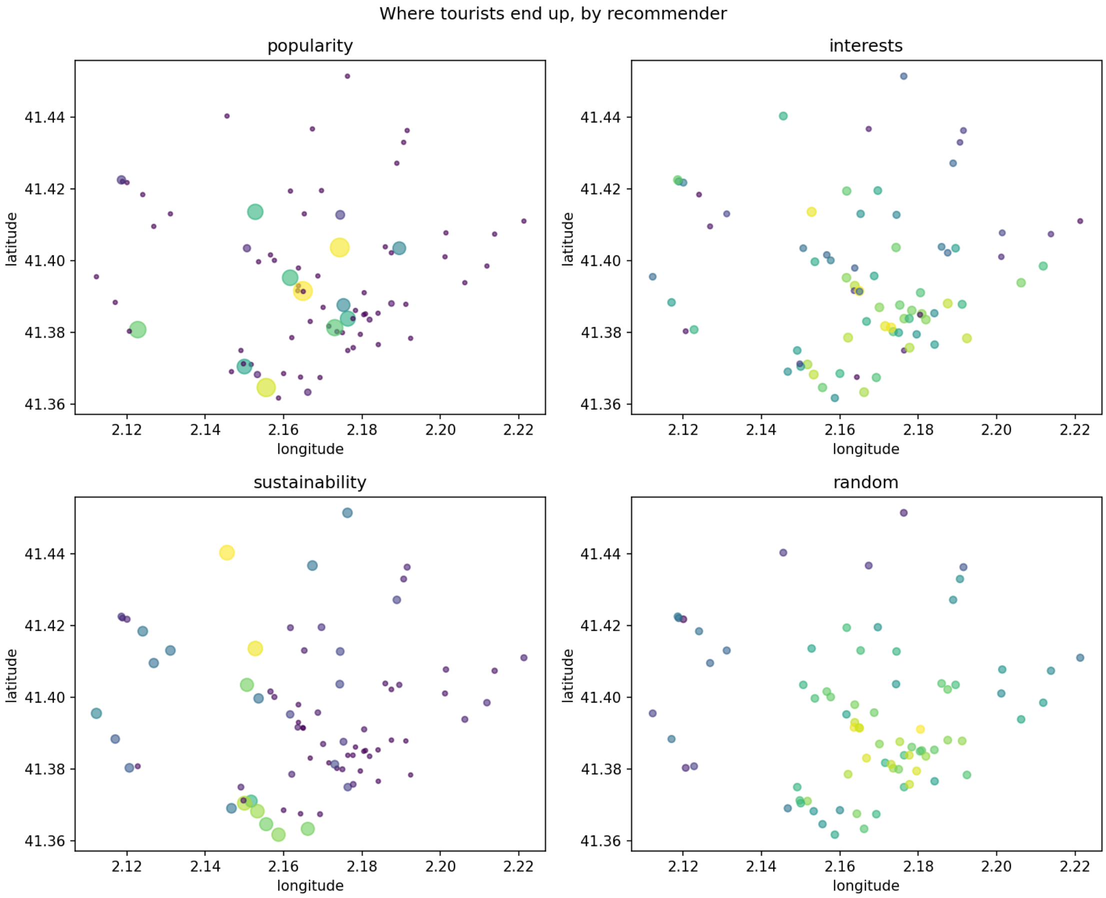

Figure 6: Where tourists end up under each recommender. Marker size and colour grow with the number of visits. Popularity puts many visitors on a few central places, the sustainability-aware recommender moves larger markers toward the edges of the city.

6.4 Sustainability of the visited places

The last group of metrics looks at the quality of the places that people actually reached, and Figure 7 summarises it. The sustainability-aware recommender is the best on the sustainability criteria that it was built for. With this strategy, visits reach places with the highest average low-pressure score (0.86 against 0.53 for popularity), the highest environmental score (0.25 against 0.08), and the highest local-economy score (0.47 against 0.28). It also keeps the cultural and accessibility scores high. These results agree with the design, because these are the criteria that the recommender optimises, and they show that the optimisation still works after tourists make their choices, even when crowding and distance are part of the decision.

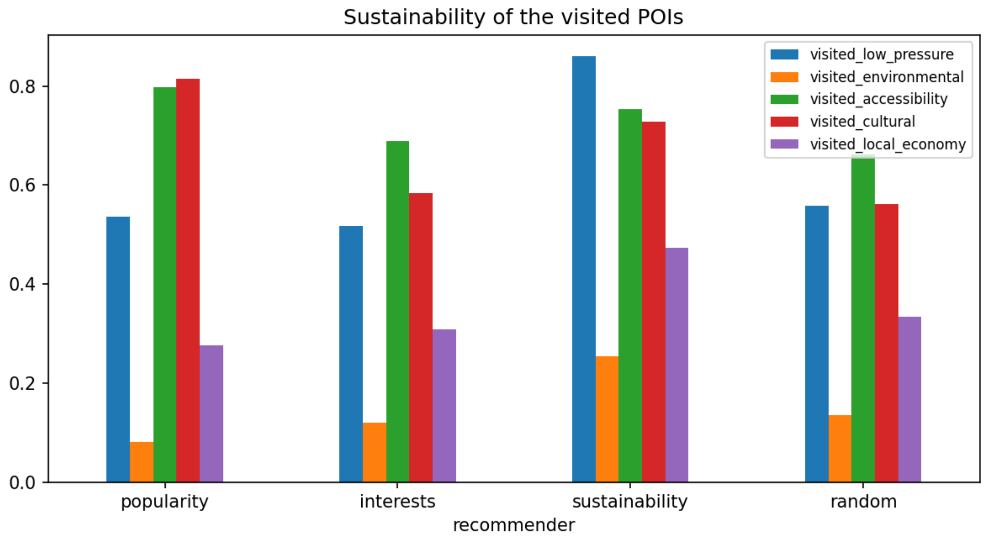

Figure 7: Average sustainability scores of the visited places, weighted by visits. The sustainability-aware recommender is the best on the criteria it optimises.

Novelty and coverage complete the picture in the same direction. The popularity recommender has the lowest novelty (4.58) and a coverage of only 0.13, which means that it depends on a small fixed group of famous places. The sustainability-aware recommender, however, reaches a coverage of 0.97 and a high novelty of 8.62, close to the random baseline, so it really shows places that visitors would not see in another way.

7 Further analyses grounded in the seminars

After the main four-way comparison, we ran three smaller studies, each one connected to a specific seminar. They keep the main comparison simple, and at the same time they show that we can extend the system in the directions that the seminars suggested.

7.1 Context-aware recommendation

The context-aware seminar [7] separated pre-filtering from post-filtering, and here we use the post-filtering method. We gave every tourist a weather context and added a step that moves outdoor places to the end of the list on rainy days. Table 3 shows the effect on the share of outdoor visits made by tourists in rainy weather. Without the context step, rainy tourists still send about 29 percent of their visits to outdoor places such as parks, beaches and viewpoints. With the context step, that share falls to zero, because the outdoor places no longer reach the top of the list. The post-filter adapts the recommendation to the weather without changing the basic ranking model, and this is the point that the seminar made about post-filtering.

Table 3: Share of visits to outdoor POIs among rainy-weather tourists.

Recommender	Rainy-day outdoor visit share
Sustainability	0.289
Sustainability + context	0.000

7.2 Tourist segmentation

The tourism seminar [4] grouped tourists into segments before it studied their behaviour. We grouped the 5,000 tourists with k-means into five segments, and we named each segment by the category that its members like most. Table 4 and Figure 8 show, for each segment, its size and two results under the sustainability recommender, the precision of the recommendations and the average low-pressure score of the visited places. The recommender does not treat every segment in the same way. Tourists who like architecture or parks receive the most accurate recommendations, near 0.79 to 0.81, while the large museum segment receives the least accurate, around 0.46, because museums compete with many sustainable options. The low-pressure score stays high in all segments, between 0.77 and 0.93, so the recommender still guides each segment toward calmer districts. We also added the personalisation metric from the same work, shown earlier in Table 1. It is zero for the popularity recommender, which gives everyone the same list, and high for the other three, which adapt the list to each tourist.

Table 4: Per-segment outcomes under the sustainability recommender.

Segment (dominant interest)	Size	Precision	Visited low-pressure
museum	2044	0.456	0.874
market	768	0.552	0.847
square landmark	736	0.595	0.774
park / nature	736	0.786	0.929
architecture	716	0.814	0.888

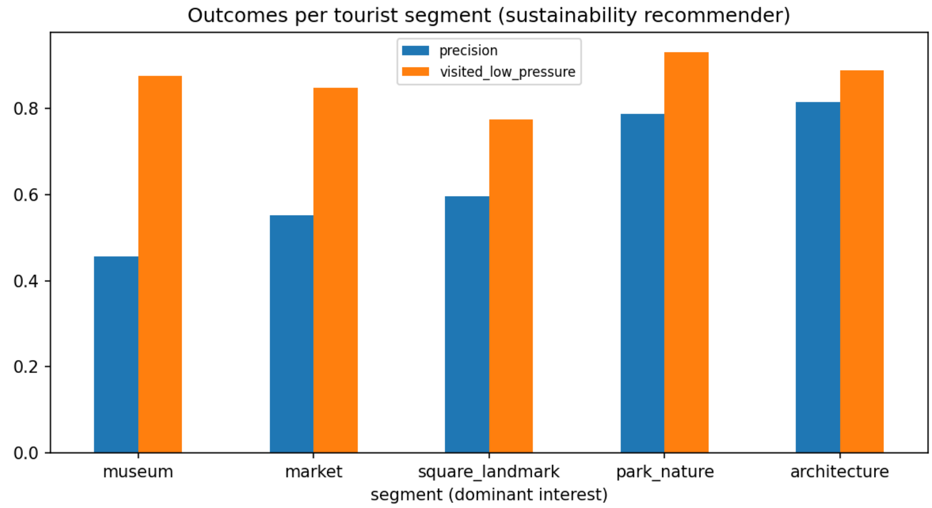

Figure 8: Precision and visited low-pressure score per tourist segment, under the sustainability recommender.

7.3 Choice of aggregation method, and explanations

The multi-criteria seminar [6] showed both ELECTRE and a simpler weighted average. We ran the sustainability recommender with each method on the same population, and we compared them in Table 5. The two methods agree on about 61 percent of their top ten places, but they are different in the other places. The weighted average gets a much higher precision (0.88 against 0.59) and a slightly lower concentration, because it respects the large weight that the profile gives to the visitor's interest. ELECTRE-III, on the other hand, sends more visitors to low-pressure districts (0.64 against 0.49), and this reason is part of the method, not random. ELECTRE compares places criterion by criterion, so the five sustainability criteria together can outrank a single preference criterion, even when that one criterion has the largest individual weight. This is important in practice, because it shows that the popular outranking method is not automatically the better choice, and that a simpler weighted average is a good alternative for this task.

Table 5: ELECTRE-III against a weighted average, same criteria and weights.

Metric	ELECTRE-III	Weighted average
Precision@10	0.591	0.878
Top-popular visit share	0.151	0.102
District visit Gini	0.432	0.270
Low-pressure district share	0.641	0.489
Top-10 overlap	0.609	

The trustworthy systems seminar [8] asked for recommendations that we can explain, and the knowledge-graph seminar [9] based those explanations on clear reasoning. Because our recommender scores each place on named criteria, we can show every suggestion as a set of criterion values. Three examples in Table 6 show this clearly. The museum lover receives the National Art Museum, which matches the interest and also scores well on most sustainability criteria, so interest and sustainability agree. The religious-site lover instead receives Parc del Laberint d'Horta, a quiet historic garden in a low-pressure district. Here the interest match is low, only 0.18, but the place has top scores on low pressure, environment and local economy, so the recommender exchanges interest for sustainability. This is the redistribution that the system wants to produce, and the criterion values show the reason to the user.

Table 6: Criterion values behind the top recommendation for three example tourists.

Tourist	Top recommendation	Pref.	Low-pr.	Env.	Acc.	Cult.	Local
museum lover	National Art Museum (museum)	0.86	0.94	0.00	1.00	1.00	0.35
religious-site lover	Parc del Laberint d'Horta (park)	0.18	0.99	1.00	0.50	0.70	0.87
nature lover	Parc del Laberint d'Horta (park)	0.86	0.99	1.00	0.50	0.70	0.87

8 Discussion

8.1 What the simulation shows

The results support a careful version of the seminar's claim. A sustainability-aware recommender can reduce overcrowding and can spread visitors more evenly across a city, and it does this while it keeps the recommendation quality much higher than a popularity baseline. The trade-off is real, and we should not hide it. The interests recommender is still the most accurate, and the sustainability-aware recommender gives up part of that accuracy to get a better distribution and better sustainability scores. A city that worries about overtourism would probably accept this exchange, but a purely commercial product might not. The decision is a policy choice, and the value of the simulator is that it gives numbers for both options.

9 Conclusion

Our goal was to test whether a sustainability-aware recommender would really improve the management of tourists in a city, and not only look good in theory. With an agent-based simulation of 5,000 tourists over 77 real points of interest in Barcelona, we compared this recommender against popularity, interests and random baselines. The sustainability-aware recommender reduced the share of visits to the most popular places from two thirds to about one sixth, reduced the number of overcrowded places, and raised the share of visits to calmer districts from less than a quarter to almost two thirds. It did this while it kept the recommendation quality clearly above the popularity baseline, but below the recommender that uses only interests. Within the limits of a simulated population, the evidence supports the seminar's idea, when we consider sustainability together with preference, tourists move away from the crowded centre and toward the rest of the city, with a cost in accuracy that a city worried about overtourism would probably accept.

References

[1] Basílica de la Sagrada Família, *La sagrada família received 4,707,367 visitors in 2023*, https://sagradafamilia.org, Accessed 2026-06-14, 2024.

[2] Barcelona de Serveis Municipals, *Park güell closes 2023 with 4.4 million visitors*, https://bsmsa.cat, Accessed 2026-06-14, 2024.

[3] Ajuntament de Barcelona, Oficina Municipal de Dades, *Observatori de districtes 2024: Tourist accommodation by district*, https://bcnroc.ajuntament.barcelona.cat, Accessed 2026-06-14, 2024.

[4] Orama, J., *Use of artificial intelligence techniques in the recommendation of points of interest*, URV AI Seminar 2026, Advanced Topics on Recommender Systems, Barcelona Supercomputing Centre, 2026.

[5] Moreno, A., *Introduction to recommender systems*, URV AI Seminar 2026, Advanced Topics on Recommender Systems, Universitat Rovira i Virgili, 2026.

[6] Valls, A., *Multi-criteria recommender systems*, URV AI Seminar 2026, Advanced Topics on Recommender Systems, Universitat Rovira i Virgili, 2026.

[7] Bellogín, A. and Cantador, I., *Context-aware recommender systems*, URV AI Seminar 2026, Advanced Topics on Recommender Systems, Universidad Autónoma de Madrid, 2026.

[8] Ricci, F., *Trustworthy recommender systems*, URV AI Seminar 2026, Advanced Topics on Recommender Systems, Free University of Bozen-Bolzano, 2026.

[9] Boratto, L., *Knowledge graphs in recommender systems*, URV AI Seminar 2026, Advanced Topics on Recommender Systems, University of Cagliari, 2026.

[10] Wikimedia Foundation, *Wikidata*, https://www.wikidata.org, Coordinates (P625) and heritage designation (P1435). Accessed 2026-06-14, 2026.

[11] OpenStreetMap contributors, *Openstreetmap*, https://www.openstreetmap.org, Accessibility and fee tags via the Overpass API. Accessed 2026-06-14, 2026.

[12] Ajuntament de Barcelona, Departament d'Estadística, *Anuari estadístic de la ciutat de barcelona 2024*, https://bcnroc.ajuntament.barcelona.cat, District areas, table 1.4. Accessed 2026-06-14, 2024.

[13] Observatori del Turisme a Barcelona, *Tourism statistics for the city and region of barcelona*, https://observatoriturisme.barcelona/en/, Accessed 2026-06-14, 2026.

[14] Pereira, V., *Pydecision: A python library for multi-criteria decision analysis*, https://github.com/Valdecy/pyDecision, 2024.

[15] Project Mesa Team, *Mesa: Agent-based modeling in python*, https://github.com/projectmesa/mesa, Version 3, 2025.